

* Python programming is a high-level programming language, it is easy to learn language used for web developing data science, AI ML automation and more.

Features of python programming:-

- interpreted language (no need to compiler)
- Simple english-like syntax
- cross-platform (windows / Linux, mac)
- open source.

How many modes executed in Python?

python programming language can be executed in two modes.

1. Interactive mode:- In this mode, lines of code are directly written and executed in the python interpreter shell, which instantaneously.

2. Normal (or) Script mode:- In this mode the source code is saved to a file with py extension and the file is executed by the python interpreter.

<u>compiler:-</u>	<u>interpreter:-</u>
→ translates the entire program at once into machine code.	→ translates and executes one line at a time.
→ After compilation you get an executable file.	→ No separate executable file is created.
→ Runs faster because translation happens only once.	→ Runs slower because it translates while executing
→ shows all errors after compiling Ex:- C, C++	→ Shows errors immediately (line by line) example:- python, JavaScript

Applications of python programming:-

Python is used for:-

- i) web Development
- ii) Data science & data Analysis
- iii) machine learning & AI
- iv) Automation & scripting
- v) Game Development
- vi) Desktop Applications
- vii) Cybersecurity & ethical Hacking
- viii) Internet of things
- ix) scientific computing
- x) mobile app development

Types of errors in python:-

- * syntax errors
- * Runtime errors
- * logical errors; these errors occur when a program compiles and runs but doesn't produce.

* Different functions of print

The print() function is one of the most basic and important functions in python. it is used to display output on the screen

```
print("addition of 'a' and 'b' = 'a+b'")
```

```
print("addition of %d and %d = %d" % (a, b, a+b))
```

```
print("addition of { } and { } = { }".format(a, b, a+b))
```

```
print(f"addition of {a} and {b} = {a+b}")
```


assignment statement:-

1. Simple assignment

* Assigns a single value to a variable.

Ex:- $x = 10$ or $\text{name} = \text{"sai"}$

x store 10 name store "sai"

2. multiple assignment:-

multiple variables, multiple values

Ex:- $a, b, c = 1, 2, 3$

~~Output~~ $\text{print}(f"a = \{a\})$

$\text{print}(f"b = \{b\})$

$\text{print}(f"c = \{c\})$

Output

$a = 1$

$b = 2$

$c = 3$

* Single value to multiple variables

Ex:- $x = y = z = 5$

↓

$x = 5, y = 5, z = 5$

condition statement:-

condition statements are used to make decisions in a program
they execute different blocks of code based on conditions

if statement:-

* executes code only if condition is True.

Ex:- $x = 10$

if $x > 5$:

$\text{print}("x \text{ is greater than } 5")$

if-else statement:-

executes one block if True, another if false.

Ex:- $x = 3$
 if $x > 5$:
 print("greater")
 else:
 print("smaller")

Output
smaller

if-elif-else:-

used when there are multiple conditions

Ex:- marks = 75
 if marks ≥ 90 :
 print("Grade A")
 elif marks ≥ 60 :
 print("Grade B")
 else:
 print("Grade C")

nested if statement:-

Ex:- $x = 10$
 if $x > 5$
 if $x < 20$
 print("x between 5 and 20")

short-hand (Ternary operator)

Ex:- $x = 10$
 print("Big") if $x > 5$ else print("small")